

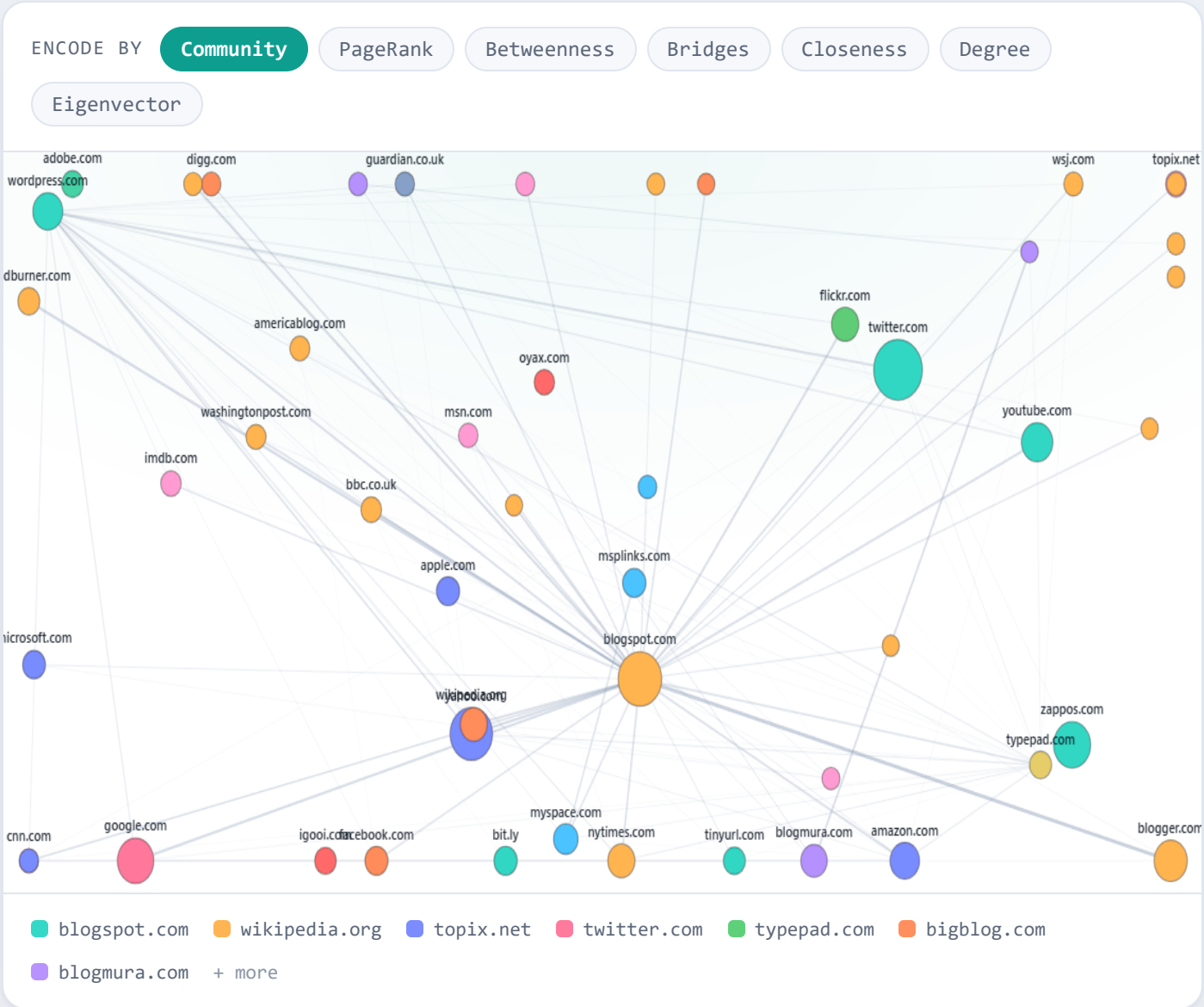
Who was central on the web in April 2009?

A **10.9 GB** MemeTracker dump — 167M lines of posts and hyperlinks — collapsed to a **domain-level link graph** and scored with seven graph algorithms, all computed in Cypher / GDS. Nodes are registered domains; a weighted edge **A→B** counts every time a post on A linked to B.

795k DOMAINS	2.21M WEIGHTED LINKS	23.55M RAW HYPERLINKS	10k CORE (DEG≥50)	3.1k COMMUNITIES
0.61 MODULARITY	524k BRIDGE EDGES			

The core link network

top 50 domains by PageRank · size & colour switch by measure



Centrality rankings

top 15 domains per measure, across all 795,268 nodes

Betweenness

sampled (2000 pivots), directed

01	blogspot.com		65.60M
02	wordpress.com		17.17M
03	google.com		6.07M
04	typepad.com		5.54M
05	flickr.com		4.96M
06	yahoo.com		2.59M
07	nytimes.com		2.57M
08	myspace.com		2.31M
09	youtube.com		2.08M
10	mahalo.com		2.00M
11	twitter.com		1.55M
12	adobe.com		1.48M
13	spiegel.de		1.34M
14	wsj.com		1.31M
15	live.com		1.30M

Bridges

bridge edges incident to the domain

01	blogspot.com		126k
02	wordpress.com		24k
03	mahalo.com		9.1k
04	typepad.com		7.5k
05	ameblo.jp		4.9k
06	kinkaa.de		4.1k
07	articlesbase.com		3.5k
08	chacha.com		3.4k
09	craigslist.org		2.9k
10	emediawire.com		2.5k
11	web-log.nl		2.0k
12	splinder.com		1.9k
13	cocolog-nifty.com		1.7k
14	exblog.jp		1.6k
15	wikia.com		1.5k

● Closeness

Wasserman-Faust, on the degree $\geq$ 50 core

01	wikipedia.org		0.654
02	blogspot.com		0.648
03	google.com		0.618
04	youtube.com		0.611
05	twitter.com		0.554
06	nytimes.com		0.547
07	wordpress.com		0.543
08	amazon.com		0.533
09	yahoo.com		0.530
10	flickr.com		0.507
11	bbc.co.uk		0.502
12	facebook.com		0.501
13	cnn.com		0.499
14	typepad.com		0.496
15	wsj.com		0.491

● Degree

weighted out-degree (Σ link weight)

01	blogspot.com		7.18M
02	wikia.com		690k
03	golivewire.com		636k
04	mashget.com		562k
05	articlesbase.com		493k
06	wordpress.com		440k
07	yunque.land.com		267k
08	ameblo.jp		236k
09	mahalo.com		231k
10	prolifeblogs.com		225k
11	feedsportal.com		212k
12	scouty.de		179k
13	rakuten.co.jp		173k
14	sportsnipe.com		166k
15	typepad.com		142k

● Eigenvector

directed, 46 iters (converged)

01	wikipedia.org		0.169
02	blogspot.com		0.168
03	google.com		0.147
04	youtube.com		0.144
05	twitter.com		0.124
06	nytimes.com		0.113
07	amazon.com		0.103
08	wordpress.com		0.098
09	yahoo.com		0.090
10	cnn.com		0.076
11	washingtonpost.co...		0.075
12	bbc.co.uk		0.075
13	wsj.com		0.074
14	facebook.com		0.072
15	msn.com		0.070

● PageRank

weighted, damping 0.85

01	twitter.com		749.42
02	blogspot.com		561.24
03	wikipedia.org		526.22
04	zappos.com		362.24
05	google.com		344.87
06	blogger.com		261.51
07	youtube.com		221.45
08	wordpress.com		192.85
09	amazon.com		184.43
10	flickr.com		142.93
11	nytimes.com		141.03
12	yahoo.com		137.91
13	blogmura.com		128.84
14	myspace.com		100.69
15	msplinks.com		83.82

Louvain communities 3,145 communities · modularity 0.61 · 12 largest shown

	blogspot.com		196k · 24.7%
	wikipedia.org		125k · 15.7%
	topix.net		79k · 10.0%
	twitter.com		58k · 7.3%
	typepad.com		40k · 5.0%
	bigblog.com		31k · 3.8%
	blogmura.com		28k · 3.5%
	wikipedia.com		27k · 3.5%
	google.de		27k · 3.4%
	flickr.com		25k · 3.1%
	igooi.com		22k · 2.8%
	myspace.com		21k · 2.6%

How each value was computed every score written back as a node property via Cypher

Betweenness `gds.betweenness`

Brandes on the directed graph, `samplingSize:2000`.
Bridges of information flow — **blogspot.com** and **wordpress.com** dominate.

Bridges `gds.bridges`

Edges whose removal disconnects the (undirected) graph. Per-domain count of incident bridges — huge for platforms that fan out to isolated blogs.

Closeness `gds.closeness`

Wasserman–Faust variant on the degree ≥ 50 core (exact all-pairs is $O(V \cdot E)$). Reference hubs like **wikipedia.org** sit closest to everything.

Degree `gds.degree`

Weighted out-degree = Σ link weight. Raw outbound volume; **blogspot.com** alone emits 7.2M links.

Eigenvector `gds.eigenvector`

Influence weighted by the influence of who links you. Converged in 46 iterations.

PageRank `gds.pageRank`

Random-surfer importance, weighted, damping 0.85. **twitter.com** tops it — 2009 was its breakout.

Louvain `gds.louvain`

Modularity community detection on the undirected weighted graph — 3,145 communities at modularity 0.61.

Source: Stanford SNAP MemeTracker (quotes_2009-04) · engine: Neo4j 2026.05 + Graph Data Science 2026.05, Docker `neo4j-gds` · graph: 795,268 domains / 2,213,774 weighted links · betweenness sampled (2,000 pivots); closeness on the degree≥50 core (10,290 nodes).